

Notes: DellaVigna and Pollet (JF, 2009)

Investor Inattention and Friday Earnings Announcements

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1 Big picture

- **Question.** Does limited investor attention affect how quickly stock prices incorporate earnings news?
- **Empirical design.** The paper compares earnings announcements made on **Friday**—when investors are plausibly more distracted because of the approaching weekend—to announcements made on other weekdays.
- **Main mechanism.** If attention is lower on Friday, stock prices should react *less immediately* to earnings surprises and then display *more delayed drift* afterward, because the neglected information is incorporated only gradually.
- **Model.** The authors build a simple equilibrium asset-pricing model with a fraction of *distracted* investors who do not observe the earnings signal when it is released. A higher distracted share attenuates the immediate response and amplifies the delayed response, while leaving the *long-run* response unchanged.
- **Main empirical result.** Friday announcements have an immediate response that is about **15% lower** and a delayed response that is about **70% higher** than announcements on other weekdays.
- **Summary statistic.** The delayed response ratio is about **60% on Friday** versus about **40% on other weekdays**. Interpreted through the model, these figures imply substantially more inattention on Friday.
- **Trading-strategy result.** A strategy that goes long the Friday post-earnings drift and short the non-Friday drift earns roughly **4% abnormal return per month**.
- **Volume result.** Trading volume also reacts less to Friday announcements. The paper finds about an **8% lower** short-run abnormal-volume response in specifications with firm fixed effects.
- **Broader contribution.** The paper provides direct evidence in favor of *underreaction due to limited attention* as an explanation for post-earnings announcement drift (PEAD).

2 Data and measurement (key objects)

2.1 Observed data sources

- **Earnings announcements and forecasts.** The core data come from **I/B/E/S** and **COMPUSTAT**. The starting sample contains quarterly earnings announcements between January 1984 and June 2006 for firms with analyst forecasts in the 90 days before the announcement.
- **Returns and volume.** Daily stock returns and trading volume come from **CRSP**.
- **Newswire validation.** To validate announcement dates, the authors hand-check a random sample of **2,487** announcements in **PR newswires** through Lexis-Nexis.

2.2 Announcement date construction

- **Why it matters.** The whole paper hinges on correctly classifying whether an announcement happened on Friday or not. Even a one-day dating error can destroy the design.

- **Starting merged sample.** The authors begin with **228,651** announcements for which I/B/E/S and COMPUSTAT dates differ by at most five calendar days.
- **Date-imputation rule.**
 1. If I/B/E/S and COMPUSTAT disagree, use the **earlier** date.
 2. If they agree *before* January 1, 1990, use the **previous trading day** because the common date often reflects Wall Street Journal publication rather than the actual announcement day.
 3. If they agree *after* January 1, 1990, use the common reported date.
- **Restriction of the sample period.** Because date accuracy is much worse before 1995, the paper restricts attention to **post-1994** announcements. In this period, the imputed date is correct more than **95%** of the time for both Friday and non-Friday announcements.

2.3 Key variables

- **Consensus forecast.** The expected earnings measure is the **median analyst forecast** in the last **30 calendar days** before the announcement, using only the analyst's most recent forecast inside that window.
- **Earnings surprise.**

$$S_{t,k} = \frac{e_{t,k} - \hat{e}_{t,k}}{P_{t-5,k}},$$

where $e_{t,k}$ is actual quarterly earnings per share, $\hat{e}_{t,k}$ is the consensus forecast, and $P_{t-5,k}$ is the stock price five trading days before the announcement.

- **Abnormal returns.** Let τ denote the announcement day. The buy-and-hold abnormal return over event window $(\tau + h, \tau + H)$ is

$$R_{t,k}^{(h,H)} = \left[\prod_{u=\tau+h}^{\tau+H} (1 + R_{u,k}) - 1 \right] - \left[\prod_{u=\tau+h}^{\tau+H} (1 + R_{u,m}) - 1 \right],$$

where $R_{u,m}$ is the market return and firm betas are estimated from a market model over days $\tau - 300$ to $\tau - 46$.

- **Abnormal volume.**

$$Av_{t,k}^{(h,H)} = \frac{1}{H - h + 1} \sum_{u=\tau+h}^{\tau+H} \log(V_{u,k}) - \frac{1}{10} \sum_{u=\tau-20}^{\tau-11} \log(V_{u,k}),$$

where $V_{u,k}$ is daily dollar volume.

- **Main return windows.**

- Immediate response: $R_{t,k}^{(0,1)}$
- Delayed response / drift: $R_{t,k}^{(2,75)}$
- Long-term response: $R_{t,k}^{(0,75)}$
- Pre-announcement window: $R_{t,k}^{(-30,-1)}$

2.4 Sample selection and descriptive facts

- **Sample cleaning.** The paper drops missing surprises, cases where earnings or forecast exceed price in absolute value, penny stocks, announcements on weekends or holidays, and extreme-return outliers.
- **Final sample.** The final analysis sample contains **143,583** announcements from January 1995 to June 2006.
- **Friday frequency.** Of these, **8,166** are Friday announcements and **135,417** are non-Friday announcements.
- **Weekday distribution.** More than **80%** of announcements occur on Tuesday, Wednesday, or Thursday, **13.8%** occur on Monday, and only **5.7%** occur on Friday.
- **Friday announcers differ somewhat.** Firms announcing on Friday have more negative earnings surprises, around **10% smaller market capitalization**, are more common earlier in the sample, and are more likely to announce in the second or third month of the quarter.
- **Homogeneous sample.** To address concern about firm heterogeneity, the paper constructs a restricted sample of firms with between **10% and 90%** of announcements on Friday. This leaves **27,130** announcements and reduces compositional differences between Friday and non-Friday subsamples.

3 Models and empirical specifications (all equations + interpretation)

3.1 Asset-pricing model with distracted investors

- **Dividend process.**

$$D_{t+1} = S + s_t + \varepsilon_{t+1},$$

where

$$s_t \sim N(0, \sigma_s^2), \quad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2).$$

The public signal s_t is the analogue of an earnings announcement.

- **Attention structure.** A fraction $1 - \mu_t$ of investors are *attentive* and observe s_t , while a fraction μ_t are *distracted* and do not observe it.
- **Day choice.** The manager observes s_t and can choose to release it on a *high-distraction* day μ_h (Friday) or a *low-distraction* day μ_ℓ (other weekdays), with

$$0 < \mu_\ell < \mu_h < 1.$$

3.2 Portfolio choice and risky-asset demand

- Investor i chooses holdings X_t^i of the risky asset to maximize mean-variance utility:

$$\max_{X_t^i} E_t^i[W_{t+1}^i] - \frac{\gamma}{2} \text{Var}_t^i(W_{t+1}^i).$$

- The demand for the risky asset is

$$X_t^i = \frac{E_t^i[P_{t+1} + D_{t+1}] - P_t(1 + R)}{\gamma \sigma_{i,t,P+D}^2}.$$

- **Interpretation.** Investors buy more of the stock when the expected payoff is high relative to today's price and buy less when payoff uncertainty is high.
- **Why attention matters.** Distracted investors do not condition on the signal, so aggregate demand responds less to news when the distracted share μ_t is higher.

3.3 Immediate, delayed, and long-term responses

The paper defines the dollar excess return

$$Z_t = P_t + D_t - (1 + R)P_{t-1}.$$

It then decomposes the reaction to the signal into three pieces.

- **Immediate response:**

$$E_t[\text{IR}_t] = Z_t - E_{t-1}[Z_t] - (D_t - E_{t-1}[D_t]).$$

- **Delayed response:**

$$E_t[\text{DR}_{t+1}] = E_t[Z_{t+1}] - E_{t-1}[Z_{t+1}].$$

- **Long-term response:** the discounted sum of the immediate and delayed responses,

$$E_t[\text{LR}_{t+1}] = E_t[\text{IR}_t] + \frac{1}{1 + R} E_t[\text{DR}_{t+1}].$$

Proposition 1 implies:

$$E_t[\text{IR}_t] = \frac{1 - b_t}{1 + R} s_t, \quad E_t[\text{DR}_{t+1}] = b_t s_t, \quad E_t[\text{LR}_{t+1}] = \frac{1}{1 + R} s_t,$$

where b_t is increasing in the distracted share μ_t .

- **Interpretation 1.** As the fraction of distracted investors rises, the immediate response gets flatter because fewer traders incorporate the signal right away.
- **Interpretation 2.** The delayed response gets larger because the neglected information is incorporated later.
- **Interpretation 3.** The total long-run reaction is unchanged. Distraction affects *timing* of price discovery, not the total amount of information eventually incorporated.

3.4 Delayed response ratio

To summarize how much of the total response is delayed, the paper defines a delayed response ratio:

$$\text{DRR}_t = \frac{E[\text{DR}_t \mid s_A, \mu_t] - E[\text{DR}_t \mid s_B, \mu_t]}{E[\text{LR}_t \mid s_A, \mu_t] - E[\text{LR}_t \mid s_B, \mu_t]} = b_t(1 + R).$$

- **Key implication.** DRR_t is increasing in μ_t , the share of distracted investors.
- **Lower bound.** The model implies

$$\mu_t \geq \text{DRR}_t.$$

- **Interpretation.** If a larger fraction of the overall stock-price adjustment shows up in the drift phase, then more investors must have ignored the information initially.

3.5 Manager timing of announcements

The paper allows the manager to choose whether to release the signal on a high-distraction or low-distraction day.

- **Long-term managers.** Managers who maximize long-run value are indifferent between Friday and other weekdays because the long-run response is independent of distraction.
- **Short-term managers.** Managers who care about the current price choose the high-distraction day when the signal is sufficiently bad:

$$\mu_t = \mu_h \quad \text{whenever} \quad s_t < -\gamma\sigma_s^2.$$

- **Interpretation.** Short-term managers strategically dump bad news on Friday because it triggers a weaker immediate price reaction.

3.6 Quantile event-study specification

For the extreme-quantile analysis, the paper estimates:

$$R_{t,k}^{(h,H)} = \alpha + \phi_{TB}d_{t,k}^{\text{top}} + \phi_{FB}d_{t,k}^{\text{top}}d_{t,k}^F + \beta d_{t,k}^F + \Gamma_0 X_{t,k} + \Gamma_1 d_{t,k}^{\text{top}} X_{t,k} + \varepsilon_{t,k}.$$

- $d_{t,k}^{\text{top}} = 1$ for the top earnings-surprise group and 0 for the bottom group.
- $d_{t,k}^F = 1$ if the announcement is on Friday.
- ϕ_{TB} measures the top-minus-bottom response for non-Friday announcements.
- ϕ_{FB} is the Friday differential relative to non-Friday announcements.
- **Prediction.** If Friday generates more distraction, then

$$\phi_{FB} < 0 \quad \text{for } R^{(0,1)}, \quad \phi_{FB} > 0 \quad \text{for } R^{(2,75)}.$$

3.7 All-announcement regression

To use the full sample rather than only the extremes, the paper estimates:

$$R_{t,k}^{(h,H)} = \alpha + \beta S_{t,k} + \beta_F S_{t,k} d_{t,k}^F + \psi d_{t,k}^F + \Gamma_0 X_{t,k} + \Gamma_1 S_{t,k} X_{t,k} + \varepsilon_{t,k}.$$

- **Interpretation.** The coefficient β is the response of returns to a one-quantile increase in earnings surprise on non-Friday announcements.
- β_F captures whether the slope is flatter or steeper on Friday.
- **Prediction.** $\beta_F < 0$ in the short run and $\beta_F > 0$ in the delayed window if Friday increases distraction.

3.8 Volume specification

For abnormal volume in the short window, the paper estimates

$$Av_{t,k}^{(0,1)} = \alpha + \sum_{j \neq 6} \alpha_j d_{t,k}^j + \phi_F d_{t,k}^F + \Gamma X_{t,k} + \varepsilon_{t,k}.$$

- **Interpretation.** If lower attention on Friday reduces trading on the information, then $\phi_F < 0$.
- **Why this is useful.** The return results could reflect either inattention or higher disagreement. Lower abnormal volume supports inattention; higher volume would instead point toward disagreement.

3.9 Portfolio strategy

- **Non-Friday drift portfolio.** In month t , go long firms that announced on non-Friday in month $t - 1$ with top earnings surprises and short firms with bottom earnings surprises.
- **Friday drift portfolio.** Do the same but only using Friday announcements from month $t - 1$.
- **Differential drift portfolio.**

$$R_t^{DNF} = R_t^{FD} - R_t^{NF}.$$

- **Interpretation.** This is a directly investable measure of whether Friday announcements generate more PEAD than non-Friday announcements.

4 Identification and interpretation (what makes the evidence credible)

4.1 Why Friday is a useful proxy for attention

- The core identifying idea is simple: Friday is just before the weekend, when investors are more likely to be distracted from work-related information processing.
- The paper does *not* claim that Friday changes the information content of earnings announcements. The claim is that Friday changes the *speed* with which the same kind of information is processed.

4.2 Date measurement is central

- A miscoded event date can mechanically generate fake Friday effects.
- The paper directly addresses this by validating dates against PR newswires and restricting the analysis to the post-1994 period where date accuracy exceeds 95%.
- This is important because the empirical design would be much weaker without precise announcement-day classification.

4.3 Why the Friday effect is not obviously compositional

- Friday announcements differ somewhat from non-Friday announcements in size, timing within the quarter, and average surprise.
- The paper therefore adds controls for year, month, market capitalization, and earnings-surprise volatility.
- It also studies the **homogeneous sample** of firms that sometimes announce on Friday and sometimes on other days, reducing the concern that a completely different set of firms drives the result.
- In the volume analysis, the paper further includes **firm fixed effects**.

4.4 Alternative explanations ruled out

- **Mechanical processing-time differences.** Friday announcements are not more likely to be after-market-close in this sample; if anything, the extra weekend time should make underreaction *less* likely, not more.
- **Pre-announcement leakage or warnings.** Figure 1(c) shows that pre-announcement returns in the window $(-30, -1)$ are similar for Friday and non-Friday announcements, conditional on earnings-surprise quantiles.
- **Announcement delay.** Late announcements could shift some reaction before day 0, but prior work finds similar information responses for on-time and late announcements, and the paper again finds no lower pre-announcement returns for Friday announcements after conditioning on surprise quantiles.
- **Higher disagreement rather than lower attention.** If Friday announcements created more disagreement, abnormal volume should be *higher*. Instead, it is lower.
- **Task overload on Monday.** If investors were simply overloaded by accumulated information on Monday, market-wide volume should be high. The paper notes instead that aggregate volume is **10% lower** on Monday.

4.5 Relation to the PEAD literature

- The paper speaks directly to the debate over why post-earnings announcement drift exists.

- Many behavioral stories can explain PEAD, but the Friday comparison is especially informative because only **underreaction caused by limited attention** predicts *more drift when attention is lower*.
- That is why the Friday-vs.-other-days comparison is not just another PEAD fact; it is evidence about *mechanism*.

5 Tables and figures

5.1 Table I and Figure 1: sample composition and baseline response curves

Table I
Summary Statistics

Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. The earnings surprise for a quarterly announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Announcements made on Saturday or Sunday are excluded from the sample. Panel A presents the distribution of earnings announcements by weekday. Columns 1 to 5 of Panel B present summary statistics with standard deviations in parentheses. Column 6 of Panel B presents the difference between columns 1 and 2, column 6 of Panel B presents the difference between columns 4 and 5, and the standard errors for these differences are in parentheses (* significant at 10%, ** significant at 5%, *** significant at 1%). The Homogeneous Sample refers to announcements by companies that have announcements on Friday and on other weekdays at least 10% of the time. Panel C presents the average earnings surprise by quantile for announcements made on Friday and on other weekdays. Quantiles 1 to 5 contain announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain announcements for five quintiles of positive earnings surprises. Quantile 6 contains announcements with an earnings surprise equal to zero. Since the number of negative earnings surprises, positive earnings surprises, and surprises equal to zero are unequal, the number of observations in each quantile differs. The breakpoints for the quantiles are determined separately for each year.

Panel A: Distribution of Earnings Announcements by Day of the Week						
	All	Monday	Tuesday	Wednesday	Thursday	Friday
Number	143,583	19,861	36,358	36,822	42,376	8,166
Fraction	1.0000	0.1383	0.2532	0.2565	0.2951	0.0569

(continued)

Table I—Continued

Panel B: Differences between Announcements on Friday and Other Weekdays

	Baseline Sample			Homogeneous Sample		
	Friday (1)	Non-Friday (2)	Difference (3)	Friday (4)	Non-Friday (5)	Difference (6)
Earnings surprise	-0.0039 (0.0353)	-0.0009 (0.0231)	-0.0030 (0.0004)***	-0.0043 (0.0370)	-0.0016 (0.0256)	-0.0027 (0.0005)***
Market Cap (\$M)	3,295 (17,599)	3,564 (15,264)	-269 (199)	3,270 (18,717)	2,820 (14,755)	639 (264)**
Year	1999.52 (3.30)	2000.05 (3.29)	-0.53 (0.0382)***	1999.49 (3.34)	1999.55 (3.29)	-0.16 (0.0489)***
Month 1 in Quarter	0.5797 (0.4938)	0.6059 (0.4886)	-0.0262 (0.0056)***	0.5906 (0.4916)	0.5909 (0.4917)	-0.0003 (0.0072)
Month 2 in Quarter	0.3166 (0.4052)	0.3114 (0.4631)	0.0052 (0.0052)	0.3146 (0.4644)	0.3174 (0.4655)	-0.0028 (0.0086)
Month 3 in Quarter	0.1037 (0.3049)	0.0867 (0.2754)	0.0210 (0.0034)***	0.0948 (0.2930)	0.0917 (0.2886)	0.0031 (0.0042)
N	N = 8,166	N = 135,417	N = 143,583	N = 5,896	N = 21,234	N = 27,130

Panel C: Average Surprise by Earnings Surprise Quantile

	Quantile	1	2	3	4	5	6	7	8	9	10	11
		Low					Surprise = 0					High
Friday	Average	-0.0523	-0.0065	-0.0025	-0.0011	-0.0004	0.0000	0.0003	0.0007	0.0014	0.0028	0.0152
	N	833	763	627	585	480	995	628	686	737	846	896
Other Days	Average	-0.0454	-0.0064	-0.0025	-0.0011	-0.0004	0.0000	0.0003	0.0007	0.0014	0.0028	0.0143
	N	7,085	7,582	7,587	7,589	7,576	20,109	15,484	15,473	15,475	15,475	15,472

Read:

- Table I establishes the key sample facts. Friday announcements are rare (**5.7%**), while more than 80% occur Tuesday–Thursday.
- Friday announcements tend to come from slightly smaller firms and contain more negative surprises, so the raw Friday/non-Friday comparison is not enough on its own; this is why the paper relies heavily on controls and robustness checks.
- Figure 1(a) is the first visual headline: the immediate response of returns to earnings surprises is visibly flatter on Friday.
- Figure 1(b) shows the mirror image: the delayed response is stronger on Friday.
- Figure 1(c) is a key credibility figure. Pre-announcement returns are very similar for Friday and non-Friday announcements conditional on the earnings-surprise quantile, which weakens the leakage/warning story.
- Figure 1(d) shows that the relation between the *actual* earnings surprise and the immediate return is nonlinear and S-shaped. This justifies the paper’s use of surprise *quantiles* rather than a simple linear surprise specification.

5.2 Table II and Figure 2: quantile event study and delayed response ratio

Read:

- Table II.A shows the immediate underreaction. With top-two versus bottom-two quantiles, non-Friday announcements have a top-minus-bottom response of **5.82%**, and Friday announcements are associated with a **15.8% lower** immediate response.

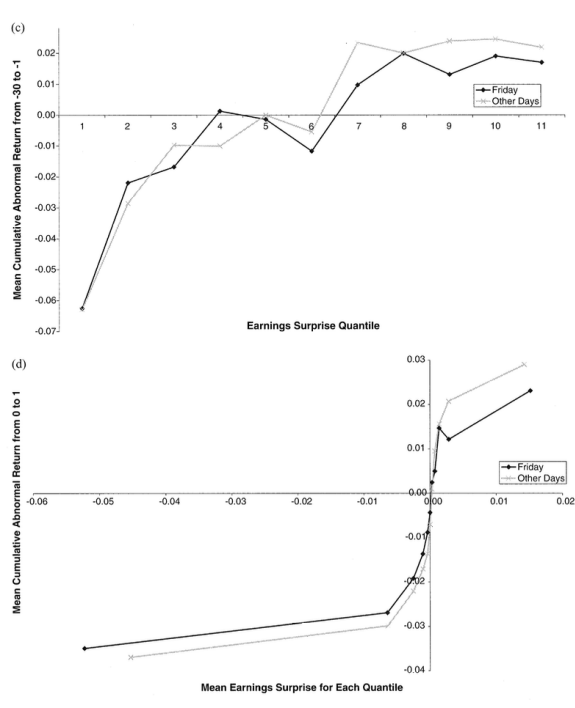


Figure 1c and d. Pre-announcement behavior before earnings surprises and nonlinear form of the immediate response to earnings surprises. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S data file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quantiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quantiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. The breakpoints for the quantiles are determined separately for each year. Announcements made on Saturday or Sunday are excluded from the sample.

Table II—Continued

[illegible]

- Table II.B shows the delayed response. Non-Friday announcements have a drift of about **5.14%** in the simple specification. Friday announcements have an additional delayed response of **4.62 percentage points**, an **89.8%** increase in the raw specification; in the controlled top-two/bottom-two specification, Friday drift is **77.2%** larger.
- Table II.C shows that the long-run response is similar or somewhat larger on Friday. This is exactly what the model predicts: attention mainly changes *timing*, not the eventual total incorporation of information.
- Table II.D summarizes the magnitude of delay. For non-Friday announcements, the delayed response ratio is around **0.42**; for Friday announcements, it is around **0.62**. Across specifications, the paper summarizes these as **40–45%** delayed for non-Friday and **54–62%** delayed for Friday.
- Figure 2 is important for timing. The extra Friday drift does not appear immediately after the weekend. Instead, it builds up after about **30 trading days** and continues to increase over the next two months. This pattern fits a slow correction process rather than a simple Monday catch-up.

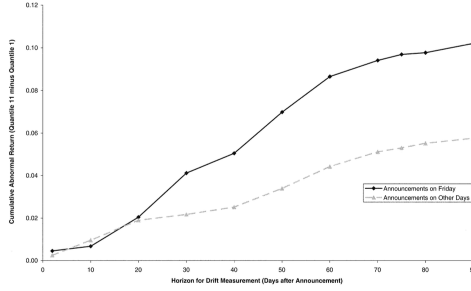


Figure 2. Performance of drift at different horizons. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1985 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. The breakpoints for the quantiles are determined separately for each year and announcements made on Saturday or Sunday are excluded from the sample. The measure for post-earnings announcement drift for horizon h is the average cumulative abnormal return from day 2 to day h for quantile 11 minus the average cumulative abnormal return from day 2 to day h quantile 1.

Table III
Stock Price Response to Earnings Announcements for All Announcements
Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1985 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. Since the number of negative earnings surprises, positive earnings surprises, and surprises equal to zero are unequal, the number of observations in each quantile differ. In Columns 3 and 6 we follow a similar procedure but sort announcements into surprise deciles instead of quantiles. The breakpoints for the quantiles (deciles) are determined separately for each year. The standard set of controls includes indicators for the year of announcement, the month of announcement, and the decile of a firm's market capitalization. The controls are present in levels and interacted with the quantile (decile) variable. Announcements made on Saturday or Sunday are excluded from the sample. Estimates from OLS regressions are reported in each column. Standard errors adjusted for heteroskedasticity and clustered by day of announcement are in parentheses (* significant at 10%; ** significant at 5%; *** significant at 1%).

	Dependent Variable: Cumulative Abnormal Return in Event Time from 0 to 1			Dependent Variable: Cumulative Abnormal Return in Event Time from 2 to 75		
	(1)	(2)	(3)	(4)	(5)	(6)
Friday	0.0067 (0.0023)***	0.0044 (0.0023)*	0.0225 (0.0019)	-0.0155 (0.0086)*	-0.0122 (0.0086)	-0.0098 (0.0069)
Earnings Surprise Quantile	0.0065 (0.0001)***	0.0047 (0.0004)***		0.0048 (0.0003)***	0.0069 (0.0015)***	
(Earnings Surprise Quantile) × Friday	-0.0011 (0.0003)***	-0.0008 (0.0003)***		0.0027 (0.0011)**	0.0023 (0.0011)**	
Earnings Surprise Decile			0.0051 (0.0004)***			0.0087 (0.0015)***
(Earnings Surprise Decile) × Friday			-0.0008 (0.0003)***			0.0024 (0.0011)**
Standard Controls (interacted)		X	X		X	X
R ²	0.0582	0.0619	0.0600	0.0028	0.0206	0.0205
N	N = 143,580	N = 143,580	N = 143,580	N = 143,580	N = 143,580	N = 143,580

5.3 Table III and Table IV / Figure 3: all-announcement regressions and trading strategy

Read:

- Table III confirms that the results are not driven only by the extreme surprises. Using all announcements, the short-run response is **16.9% flatter** on Friday, and the delayed response is **56.2% larger**.
- This is reassuring because it shows the Friday effect is a general pattern in the data rather than an artifact of looking only at the tails.
- Figure 3 shows the yearly average returns of the Friday and non-Friday drift portfolios. The Friday drift portfolio is more volatile because it contains far fewer stocks, but it beats the non-Friday drift portfolio in **10 of 12** years.
- Table IV converts the reduced-form evidence into an implementable trading strategy. The differential Friday-drift portfolio earns a monthly abnormal return of **3.84%** in the baseline specification.

- The matched-surprise version earns **5.84%**, which is useful because it addresses the concern that Friday announcements may simply be more extreme.
- Broader portfolio definitions still generate significant returns: **2.14%** when using top-two/bottom-two groups and **2.77%** with decile sorting.
- The paper interprets these portfolio profits as economically large evidence that Friday announcements generate more underreaction than non-Friday announcements.

[illegible]

Table IV
Performance of Post-Earnings Announcement Drift Portfolios

[illegible]Table IV—Continued
Monthly Returns on the Zero-Investment Portfolio

5.4 Figure 4 and Table V: volume response

- Figure 4 shows that for non-Friday announcements, abnormal volume rises to about **47%** on the announcement day and **61%** on the next trading day, then decays slowly.
- For Friday announcements, volume is somewhat higher on the announcement day but **43% lower** on the next trading day, so the combined immediate response window has clearly less trading.
- Table V formalizes this. Without controls, Friday announcements have a short-run abnormal-volume response that is **20.5% lower**. With firm fixed effects, the attenuation is still about **8.1%**.
- The volume result is crucial for interpretation. Lower volume supports limited attention; it is difficult to reconcile with stories based on higher disagreement or richer information release.
- Once aggregate abnormal volume is included as a control, the Friday coefficient becomes insignificant, but the paper argues this is still consistent with the inattention hypothesis because inattention should depress both stock-specific and market-wide trading on Friday and Monday.

- **Method.** The paper combines a simple equilibrium model of distracted investors with reduced-form event-study evidence comparing Friday and non-Friday earnings announcements.

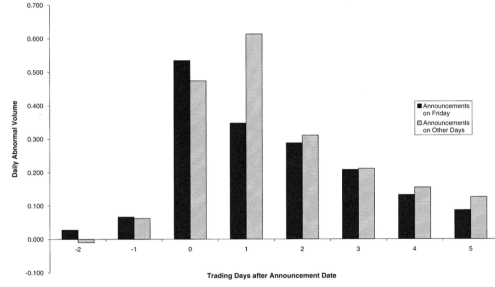


Figure 4. Abnormal volume around earnings announcement date. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The abnormal volume for each stock is the average log volume on the day of and the day after the announcement, divided by the average log volume for the period -20 to -11 in event time (10 trading days). Announcements made on Saturday or Sunday are excluded from the sample.

Table V
Short-Term Volume Response to an Earnings Announcement

Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The abnormal volume for each stock is the average log volume on the day of and the day after the announcement, divided by the average log volume for the period -20 to -11 in event time (10 trading days). The aggregate abnormal volume is computed in a similar manner using volume for the whole market. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement.

Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. Since the number of negative earnings surprises, positive earnings surprises, and surprises equal to zero are not equal, the number of observations in each quantile differ. In Column 5 we follow a similar procedure but sort announcements into surprise deciles instead of quantiles. The breakpoints for the quantiles (deciles) are determined separately for each year. Indicators for each earnings surprise quantile (decile) are included in all regressions.

The standard set of controls includes indicators for the year of announcement, the month of announcement, and the decile of a firm's market capitalization. The additional earnings surprise volatility controls in columns 4 and 5 are indicators for the decile of the company's earnings surprise standard deviation during the previous 4 years (requiring at least four observations). Announcements made on Saturday or Sunday are excluded from the sample. Estimates from OLS regressions are reported in each column. Standard errors adjusted for heteroskedasticity and clustered by day of announcement are in parentheses (* significant at 10%, ** significant at 5%, *** significant at 1%).

	Dependent Variable: Abnormal Volume in Event Time from 0 to 1					
	(1)	(2)	(3)	(4)	(5)	(6)
Friday	-0.1109 (0.0161)***	-0.1235 (0.0130)***	-0.0436 (0.0129)***	-0.0413 (0.0130)***	-0.0414 (0.0129)***	-0.0173 (0.0121)
Aggregate Abnormal Volume						0.3977 (0.0289)***
Earnings Surprise Quantiles	X	X	X	X		X
Earnings Surprise Deciles					X	
Standard Controls		X	X	X	X	X
Surprise Volatility Controls				X	X	
Company Fixed Effects			X	X	X	X
R ²	0.0072	0.0344	0.1963	0.1987	0.1984	0.2021
N	N = 143,369	N = 143,369	N = 143,369	N = 122,325	N = 122,325	N = 143,369

- **Identification.** The key design uses Friday as a proxy for lower investor attention and strengthens credibility through careful date validation, controls, a homogeneous-sample exercise, volume evidence, and multiple alternative-explanation tests.
- **Main facts.** Friday announcements have a weaker immediate price response, a stronger delayed drift, a larger delayed-response share of total adjustment, and lower abnormal trading volume.
- **Interpretation.** These patterns support the view that PEAD reflects underreaction to information caused by limited attention.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that the speed of information incorporation in stock prices depends on investor attention.
- A simple and intuitive attention shock—Friday before the weekend—materially slows the immediate reaction of prices to earnings news and increases subsequent drift.
- Because the total long-run response is not much lower, the key effect of attention is on *timing*, not on the eventual amount of information reflected in prices.

7.2 Economic intuition

- When fewer investors are paying attention, fewer trades immediately reflect the announcement.
- Prices therefore move too little at first, and some of the adjustment is postponed until investors later revisit the information.
- Lower attention should also reduce the trading volume associated with the news, and that is exactly what the paper finds.

- If managers care mainly about the short-term stock price, they have an incentive to release especially bad news on high-distraction days, which links the asset-pricing evidence to corporate disclosure timing.

7.3 Takeaway

This paper provides unusually clean evidence on *mechanism*. Many papers document post-earnings announcement drift; this one shows that drift becomes stronger exactly when attention is plausibly lower. That makes limited attention not just a vague behavioral story, but a concrete and empirically supported explanation for why earnings news is incorporated into stock prices with delay.